National Testing Agency

Question Paper Name :	B Tech 5th Apr 2026 Shift 2
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Creation Date :	2026-04-05 19:23:59
Duration :	180
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Display Marks:	Yes

B. Tech

Group Number :	1
Group Id :	6911217
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Id :	69112137
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112137
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 691121451 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let α, β be the roots of the equation $x^2 - x + p = 0$ and γ, δ be the roots the equation $x^2 - 4x + q = 0$; $p, q \in \mathbf{Z}$. If $\alpha, \beta, \gamma, \delta$ are in G.P., then $|p + q|$ equals :

Options :

6911211531. 16

6911211532. 32

6911211533. 34

6911211534. 38

Question Number : 2 Question Id : 691121452 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $z_1, z_2 \in \mathbb{C}$ be the distinct solutions of the equation $z^2+4z-(1+12i)=0$.
Then $|z_1|^2+|z_2|^2$ is equal to :

Options :

6911211535. 18

6911211536. 22

6911211537. 29

6911211538. 34

Question Number : 3 Question Id : 691121453 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $f: \mathbf{N} \to \mathbf{Z}$ is defined by

$$f(n)=\begin{vmatrix} n & -1 & -5 \\ -2n^2 & 3(2k+1) & 2k+1 \\ -3n^3 & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix}, k \in \mathbf{N},$$

and $\sum_{n=1}^{k} f(n)=98$, then k is equal to :

Options :

6911211539. 3

6911211540. 4

6911211541. 5

6911211542. 6

Question Number : 4 Question Id : 691121454 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let M be a 3×3 matrix such that

$$M\begin{pmatrix}1\\0\\0\end{pmatrix}=\begin{pmatrix}1\\2\\3\end{pmatrix}, M\begin{pmatrix}0\\1\\0\end{pmatrix}=\begin{pmatrix}0\\1\\2\end{pmatrix} \text{ and } M\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}-1\\1\\1\end{pmatrix}. \text{ If } M\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1\\7\\11\end{pmatrix}, \text{ then } x+y+z \text{ equals :}$$

Options :

6911211543. 4

6911211544. 5

6911211545. 7

6911211546. 11

Question Number : 5 Question Id : 691121455 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the sum of the first 10 terms of the series $\frac{1}{1+1^4\times 4}+\frac{2}{1+2^4\times 4}+\frac{3}{1+3^4\times 4}+\frac{4}{1+4^4\times 4}+.....$ is $\frac{m}{n}$, gcd $(m, n)=1$, then $m+n$ is equal to :

Options :

6911211547. 256

6911211548. 264

6911211549. 276

6911211550. 284

Question Number : 6 Question Id : 691121456 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A_1, A_2, A_3, ..., A_{39}$ be 39 arithmetic means between the numbers 59 and 159. Then the mean of A_{25}, A_{28}, A_{31} and A_{36} is equal to :

Options :

6911211551. 129

6911211552. 136

6911211553. 131.50

6911211554. 134

Question Number : 7 Question Id : 691121457 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The coefficient of x^2 in the expansion of $\left(2x^2+\frac{1}{x}\right)^{10}, x\neq 0$, is :

Options :

6911211555. 3240

6911211556. 3360

6911211557. 3480

6911211558. 3600

Question Number : 8 Question Id : 691121458 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The probabilities that players A and B of a team are selected for the captaincy for a tournament are 0.6 and 0.4, respectively. If A is selected the captain, the probability that the team wins the tournament is 0.8 and if B is selected the captain, the probability that the team wins the tournament is 0.7. Then the probability, that the team wins the tournament, is :

Options :

6911211559. 0.74

6911211560. 0.76

6911211561. 0.72

6911211562. 0.78

Question Number : 9 Question Id : 691121459 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A box contains 5 blue, 6 yellow and 4 red balls. The number of ways, of drawing 8 balls containing at least two balls of each colour, is :

Options :

6911211563. 4100

6911211564. 4140

6911211565. 4230

6911211566. 4290

Question Number : 10 Question Id : 691121460 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A variable X takes values 0, 0, 2, 6, 12, 20, ..., $n(n-1)$ with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3, {}^nC_4, {}^nC_5, ..., {}^nC_n$, respectively. If the mean of this data is 60, then its median is :

Options :

6911211567. 56

6911211568. 42

6911211569. 72

6911211570. 90

Question Number : 11 Question Id : 691121461 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the point P be the vertex of the parabola $y=x^2-6x+12$. If a line passing through the point P intersects the circle $x^2+y^2-2x-4y+3=0$ at the points R and S, then the maximum value of $(PR+PS)^2$ is :

Options :

6911211571. 10

6911211572. 20

6911211573. 25

6911211574. 5

Question Number : 12 Question Id : 691121462 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the directrix of the parabola P : $y^2=8x$, cut x-axis at the point A. Let B (α, β), $\alpha > 1$, be a point on P such that the slope of AB is 3/5. If BC is a focal chord of P, then six times the area of ΔABC is :

Options :

6911211575. 80

6911211576. 160

6911211577. 174

6911211578. 192

Question Number : 13 Question Id : 691121463 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the eccentricity e of a hyperbola satisfy the equation $6e^2-11e+3=0$. If the foci of the hyperbola are (3, 5) and (3, -4), then the length of its latus rectum is :

Options :

6911211579. $11/3$

6911211580. $17/3$

6911211581. $15/2$

6911211582. $17/2$

Question Number : 14 Question Id : 691121464 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let a triangle PQR be such that P and Q lie on the line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ and are at a distance of 6 units from R (1, 2, 3). If (α, β, γ) is the centroid of ΔPQR, then $\alpha+\beta+\gamma$ is equal to :

Options :

6911211583. 4

6911211584. 5

6911211585. 6

6911211586. 8

Question Number : 15 Question Id : 691121465 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the distance of the point (a, 2, 5) from the image of the point (1, 2, 7) in the line $\frac{x}{1}=\frac{y-1}{1}=\frac{z-2}{2}$ is 4, then the sum of all possible values of a is equal to :

Options :

6911211587. 11

6911211588. 9

6911211589. 6

6911211590. 4

Question Number : 16 Question Id : 691121466 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let O be the origin, $\vec{OP}=\vec{a}$ and $\vec{OQ}=\vec{b}$. If R is the point on $\vec{OP}$ such that $\vec{OP}=5\vec{OR}$, and M is the point such that $\vec{OQ}=5\vec{RM}$, then $\vec{PM}$ is equal to :

Options :

6911211591. $\frac{1}{5}\left(\vec{a}-4\vec{b}\right)$

6911211592. $\frac{1}{5}\left(\vec{b}-4\vec{a}\right)$

6911211593. $\frac{1}{5}\left(-\vec{a}+4\vec{b}\right)$

6911211594. $\frac{1}{5}\left(-\vec{b}+4\vec{a}\right)$

Question Number : 17 Question Id : 691121467 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f(x)=\lim\limits_{y\to 0}\frac{(1-\cos(xy))\tan(xy)}{y^3}$. Then the number of solutions of the equation $f(x)=\sin x$, $x\in\mathbf{R}$ is :

Options :

6911211595. 0

6911211596. 2

6911211597. 3

6911211598. 1

Question Number : 18 Question Id : 691121468 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $(2^{1-a}+2^{1+a})$, $f(a)$, (3^a+3^{-a}) be in A.P. and α be the minimum value of $f(a)$. Then the value of the integral $\int_{\log_e(\alpha-1)}^{\log_e(\alpha)}\frac{dx}{\left(e^{2x}-e^{-2x}\right)}$ is :

Options :

6911211599. $\frac{1}{2}\log_e\left(\frac{4}{3}\right)$

6911211600. $\frac{1}{4}\log_e\left(\frac{4}{3}\right)$

6911211601. $\frac{1}{2}\log_e\left(\frac{8}{5}\right)$

6911211602. $\frac{1}{4}\log_e\left(\frac{8}{5}\right)$

Question Number : 19 Question Id : 691121469 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f : [1, \infty) \to \mathbf{R}$ be a differentiable function defined as $f(x) = \int_1^x f(t)dt + (1-x)(\log_e x - 1) + e$.

Then the value of $f(f(1))$ is :

Options :

6911211603. $(1 + e^e)$

6911211604. $(1 + e)$

6911211605. $(1 + e + e^e)$

6911211606. $1 + 2e$

Question Number : 20 Question Id : 691121470 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f(x)$ and $g(x)$ be twice differentiable functions satisfying $f''(x) = g''(x)$ for all $x \in \mathbf{R}, f'(1) = 2g'(1) = 4$ and $g(2) = 3f(2) = 9$. Then $f(25) - g(25)$ is equal to :

Options :

6911211607. 20

6911211608. 40

6911211609. -20

6911211610. -40

Mathematics Section B

Section Id :	69112138
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112138
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Id : 691121471 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $A = \{1, 4, 7\}$ and $B = \{2, 3, 8\}$. Then the number of elements, in the relation $R = \{((a_1, b_1), (a_2, b_2)) \in ((A \times B) \times (A \times B)) : a_1 + b_2 \text{ divides } a_2 + b_1\}$ is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 22 Question Id : 691121472 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

From the point $(-1, -1)$, two rays are sent making angles of $45°$ with the line $x+y=0$. These rays get reflected from the mirror $x+2y=1$. If the equations of the reflected rays are $ax+by=9$ and $cx+dy=7$, a, b, c, $d \in \mathbf{Z}$, then the value of $ad+bc$ is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 23 Question Id : 691121473 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If $S=\left\{\theta \in [-\pi, \pi] : \cos\theta \cos\frac{5\theta}{2} = \cos 7\theta \cos\frac{7\theta}{2}\right\}$, then n(S) is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 24 Question Id : 691121474 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $f: \mathbf{R} \to \mathbf{R}$ be a function such that $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x$, $x \in \mathbf{R}$. Let the maximum value of f on $\mathbf{R}$ be α. If the area of the region bounded by the curves $g(x)=x^2$ and $h(x)=\beta x^3$, $\beta > 0$, is α^2, then $30\beta^3$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 25 Question Id : 691121475 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $y=y(x)$ be the solution of the differential equation

$(\tan x)^{1/2}\, dy = \left(\sec^3 x - (\tan x)^{3/2} y\right) dx$, $0 < x < \frac{\pi}{2}$, $y\left(\frac{\pi}{4}\right) = \frac{6\sqrt{2}}{5}$. If $y\left(\frac{\pi}{3}\right) = \frac{4}{5}\alpha$, then α^4 equals _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes